



Figure 4: Update settings

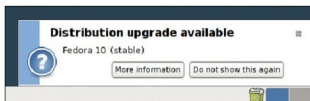


Figure 5: Notification on the availability of a new OS version



Figure 6: Upgrade to new distro version with PreUpgrade



Figure 7: PackageKit informs you of the need for additional codec installation

Classification of updates

PackageKit classifies updates under security, bug fixes and enhancements, and you can choose to selectively update

either interactively or set the preferences to automatically install them. If you are on a low-bandwidth connection, setting it to auto install security updates might be an ideal thing to do.

Environment aware—bandwidth and power management

PackageKit is aware of when you are using a mobile Net connection, and does not drain your bandwidth and increase your bills even if you have set it to update automatically (Figure 4).

It is also aware when you are running on battery, and it wouldn't run updates by default in this case. The option to tweak this is an advanced setting not visible via the preferences dialogue box but you can change it via a GConf key.

Distribution upgrades

With the fast pace of free and open source software updates and many community distributions like Fedora coming up with new releases virtually every six months, users are often unaware that a new release is available. They continue to use old and outdated releases, sometimes even without getting any security updates, which leads to potential security issues.

PackageKit makes the process of upgrading to a new release just a bit easier. When a new release is available, PackageKit provides a notification on your desktop itself (Figure 5).

The upgrade process is managed by the native distribution tools. In Fedora, that's PreUpgrade, which provides an online way to upgrade to a new release easily.

When a user initiates an upgrade, PackageKit downloads PreUpgrade and executes it. PreUpgrade then continues doing the actual upgrade process (Figure 6).

On-demand installation

In Fedora 10, PackageKit has a feature of adding codecs on demand. Let's suppose you click on a music file, which is encoded in a format that doesn't have support for it out-of-the-box. In most cases, previously you would get a cryptic error and you wouldn't be able to do much with it. With PackageKit, there is a *gststreamer* plug-in and you get a nice descriptive dialogue box that guides you through the process (Figure 7).

Of course, you still need an appropriate plug-in, which is available in the repository. In the case of Fedora, you would need a third party repository like RPM Fusion enabled, but PackageKit will figure out the right plug-in all by itself. On demand installation is supported just for codecs now but much more is planned for future versions. More on this later.

Service packs—offline software installation

In Linux distributions, a rich choice of the software packages and updates is usually available in a central software repository, but not everybody has a broadband